

A per-atom error model for the crystallographic likelihood: from atomic error statistics to the Rice target, its spatial constraint, and its least-squares limit

Working note

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Scope and status

This note derives the amplitude likelihood for an atomic model in which every atom carries its own error description, shows that Read’s σ_A likelihood is the uniform special case, introduces spatially smooth fields as the constraint that makes the per-atom model estimable, and shows that in the limit of a good model and strong data the target reduces to weighted least squares on intensities with inverse-variance weights and a Fisher-information closure for the atomic errors.

Experimental error is deliberately omitted throughout; it enters later as an additive variance.

Sections 1–5 follow the classical route (Luzzati, Srinivasan & Ramachandran, Read) with the per-atom generalisation made explicit. Sections 6–7 are design choices. Sections 8–10 are derived in this note and are to be treated as claims until the numerical checks in Section 11 pass. Every approximation is labelled (A1)–(A5) so that it can be tested in isolation.

Notation

Symbol	Meaning
h	Miller index (reciprocal-lattice vector), treated as a vector in reciprocal $\text{\AA}^{-1}$ where a quadratic form $h^\top U h$ appears.
s	$ h $, the reciprocal resolution $1/d$. Functions written as $f(s)$ depend on resolution only.
ε	Expected-intensity multiplicity (symmetry enhancement factor) of reflection h .
N	Number of atoms in the model.
x_j	Model position of atom j (Cartesian, $\text{\AA}$).
$f_j(s)$	Atomic scattering factor of atom j .
$T_j(h)$	Temperature factor of atom j from its modelled ADP, $T_j = \exp(-2\pi^2 h^\top U_j^{\text{ADP}} h)$. This is part of the model, not of the error.
ξ_j	Indicator that atom j is real: $\xi_j \in \{0, 1\}$, $\Pr(\xi_j = 1) = w_j$.
w_j	Presence probability of atom j (the probability that the modelled atom exists in the true structure near x_j).

Symbol	Meaning
Δ_j	Positional error of atom j when it is real: the displacement of the true atom from x_j .
U_j	3×3 covariance of Δ_j (the error covariance, $\text{\AA}^2$), distinct from the ADP U_j^{ADP} .
B_{err}	Isotropic error B-value; $U_j = (B_{\text{err}}/8\pi^2)\mathbb{I}$ in the isotropic case.
$D_j(h)$	Per-atom Luzzati factor $\mathbb{E}[e^{2\pi i h \cdot \Delta_j}] = \exp(-2\pi^2 h^\top U_j h)$.
$d_j(h)$	Per-atom effective weight $w_j D_j(h)$: the fraction of atom j 's scattering that is predictable.
g_m, y_m	Scattering factor and position of the m -th unmodelled scatterer ($m = 1, \dots, M$).
$\Sigma_{\text{miss}}(s)$	Scattering power of the unmodelled part, $\sum_m g_m^2$.
$F(h)$	True structure factor of the crystal (the quantity whose modulus is observed).
$F_c(h)$	Unweighted model structure factor, $\sum_j f_j T_j e^{2\pi i h \cdot x_j}$.
$F_{\text{eff}}(h)$	Expected true structure factor given the model and error model, $\mathbb{E}[F(h)]$.
$\Sigma_\Delta(h)$	Model-error variance, $\mathbb{E}[F(h) - F_{\text{eff}}(h)]^2$.
$\Sigma_P(s), \Sigma_N(s)$	Read's scattering power of the model atoms and of the true structure, respectively.
$D(s), \sigma_A(s)$	Read's uniform Luzzati factor and σ_A .
I_h	Observed intensity $ F(h) ^2$ (no experimental error in this note).
$D_0(s)$	Shell table: the spatial mean of d_j as a function of resolution.
$\lambda(x), \kappa(x)$	Real-space fields: log presence contrast and excess error-B, both mean-zero.
c_k, v_k	Fourier coefficient of a field at reciprocal index k , and its prior variance.
$p_j(s)$	Explained-power distribution over atoms in resolution shell s (used by the entropy regulariser).
α	Weight of the entropy regulariser.
θ	Vector of all refined model parameters (coordinates, ADPs, occupancies, scales).
J_h	Gradient $\partial F_{\text{eff}}(h) ^2/\partial\theta$.
M	Gauss–Newton normal matrix (observed Fisher information).

1 Generative model: one error description per atom

The model consists of N atoms at positions x_j with scattering $f_j(s)T_j(h)$. The true structure is treated as a random draw around the model, with the following ingredients.

- (i) Atom j is real with probability w_j ; the indicator $\xi_j \sim \text{Bernoulli}(w_j)$.
- (ii) If real, the true atom sits at $x_j + \Delta_j$ with $\Delta_j \sim \mathcal{N}(0, U_j)$.
- (iii) Scattering absent from the model is represented by M scatterers g_m at positions y_m drawn uniformly over the cell.

(A1) All ξ_j , Δ_j and y_m are mutually independent. **(A2)** Positional errors are Gaussian. **(A5)** Unmodelled scatterers are uniformly distributed (no envelope).

The true structure factor is then

$$F(h) = \sum_{j=1}^N \xi_j f_j T_j e^{2\pi i h \cdot (x_j + \Delta_j)} + \sum_{m=1}^M g_m e^{2\pi i h \cdot y_m}. \quad (1)$$

Because experimental error is omitted, the observation is $I_h = |F(h)|^2$ exactly.

2 First and second moments

Per-atom Luzzati factor. The characteristic function of the Gaussian displacement (A2) is

$$D_j(h) = \mathbb{E} \left[e^{2\pi i h \cdot \Delta_j} \right] = \exp \left(-2\pi^2 h^\top U_j h \right). \quad (2)$$

For isotropic error, $U_j = (B_{\text{err},j}/8\pi^2)\mathbb{I}$ and $D_j = \exp(-B_{\text{err},j} s^2/4)$, Luzzati's form.

Effective weight. Define $d_j(h) = w_j D_j(h)$. This is the fraction of atom j 's scattering that can be predicted: presence times phase coherence.

First moment. Taking expectations in (1) under (A1) and (A5), the unmodelled term averages to zero and

$$F_{\text{eff}}(h) \equiv \mathbb{E}[F(h)] = \sum_j d_j(h) f_j T_j e^{2\pi i h \cdot x_j}. \quad (3)$$

This is the REFMAC group sum $\sum_g D_g F_g$ with one group per atom.

Second moment. For one atom, use $\xi_j^2 = \xi_j$ and $|e^{i\phi}|^2 = 1$, so $\mathbb{E}|\xi_j e^{2\pi i h \cdot \Delta_j}|^2 = w_j$. The variance contributed by atom j is therefore $f_j^2 T_j^2 (w_j - d_j^2)$. Each unmodelled scatterer contributes g_m^2 . With independence (A1) the contributions add:

$$\Sigma_\Delta(h) \equiv \mathbb{E}|F(h) - F_{\text{eff}}(h)|^2 = \sum_j f_j^2 T_j^2 (w_j - d_j(h)^2) + \Sigma_{\text{miss}}(s). \quad (4)$$

The pair $(F_{\text{eff}}, \Sigma_\Delta)$ is the only route by which the error model enters the likelihood. Both are built from the same per-atom quantities (w_j, U_j) ; this shared origin is what will later make the error model identifiable against occupancies and ADPs.

Pseudo-variance and circularity. The quantity $\mathbb{E}[(F - F_{\text{eff}})^2]$ (no modulus) is not zero atom by atom: atom j contributes $f_j^2 T_j^2 e^{4\pi i h \cdot x_j} (w_j D_j(2h) - d_j(h)^2)$. Summed over atoms these carry the phases $e^{4\pi i h \cdot x_j}$ and, for atoms in general position, cancel to $O(\sqrt{N})$ against the $O(N)$ variance in (4). **(A4)** The pseudo-variance is neglected, i.e. the error is taken to be circular. This is the standard Wilson-type step. For centric reflections it is replaced by the exact statement that $F - F_{\text{eff}}$ is real.

3 Distribution of the true structure factor

$F - F_{\text{eff}}$ is a sum of $N + M$ independent terms, each small relative to the total. **(A3)** By the central limit theorem (Lindeberg condition) it is approximately Gaussian. With (A4) it is circular:

$$F(h) \sim \mathcal{CN}(F_{\text{eff}}(h), \varepsilon \Sigma_{\Delta}(h)) \quad (\text{acentric}), \quad (5)$$

where $\mathcal{CN}(\mu, \sigma^2)$ denotes the circular complex normal with $\mathbb{E}|F - \mu|^2 = \sigma^2$. For centric reflections

$$F(h) \sim \mathcal{N}(F_{\text{eff}}(h), \varepsilon \Sigma_{\Delta}(h)) \quad (\text{centric, } F \text{ real}). \quad (6)$$

4 The Rice-like likelihood

Integrating the unobserved phase out of (5) gives the acentric amplitude density

$$p(|F|; F_{\text{eff}}, \Sigma_{\Delta}) = \frac{2|F|}{\varepsilon \Sigma_{\Delta}} \exp\left(-\frac{|F|^2 + |F_{\text{eff}}|^2}{\varepsilon \Sigma_{\Delta}}\right) I_0\left(\frac{2|F||F_{\text{eff}}|}{\varepsilon \Sigma_{\Delta}}\right), \quad (7)$$

where I_0 is the modified Bessel function of order zero. This is the Rice distribution. From (6) the centric density is Woolfson's

$$p(|F|; F_{\text{eff}}, \Sigma_{\Delta}) = \sqrt{\frac{2}{\pi \varepsilon \Sigma_{\Delta}}} \exp\left(-\frac{|F|^2 + |F_{\text{eff}}|^2}{2\varepsilon \Sigma_{\Delta}}\right) \cosh\left(\frac{|F||F_{\text{eff}}|}{\varepsilon \Sigma_{\Delta}}\right). \quad (8)$$

In intensities, $p(I) = p(|F|)/(2|F|)$ with $|F| = \sqrt{I}$. Equivalently, $2I/(\varepsilon \Sigma_{\Delta})$ is a non-central χ^2 with two (acentric) or one (centric) degrees of freedom and non-centrality parameter $2|F_{\text{eff}}|^2/(\varepsilon \Sigma_{\Delta})$.

The refinement target is the negative log-likelihood

$$\mathcal{L}_{\text{data}} = -\sum_h \log p(I_h; F_{\text{eff}}(h), \Sigma_{\Delta}(h)). \quad (9)$$

Nothing in this section depends on the error parameters being per-atom: the Rice-like distribution for the per-atom model is the ordinary Rice distribution with F_{eff} from (3) and Σ_{Δ} from (4).

5 Check: the uniform case is Read's σ_A

Let every atom share the same error, $w_j = w$ and $U_j = (B_{\text{err}}/8\pi^2)\mathbb{K}$. Then $d_j = D(s)$ with

$$D(s) = w \exp(-B_{\text{err}} s^2/4), \quad (10)$$

so $F_{\text{eff}} = D(s) F_c$. With $\Sigma_P = \sum_j f_j^2 T_j^2$, equation (4) becomes

$$\Sigma_{\Delta} = w \Sigma_P - D^2 \Sigma_P + \Sigma_{\text{miss}} = \Sigma_N - D^2 \Sigma_P, \quad \Sigma_N \equiv w \Sigma_P + \Sigma_{\text{miss}}. \quad (11)$$

Here Σ_N is the scattering power of the true structure: the real fraction of the model plus the unmodelled part. Normalising to E -values with Σ_N gives

$$\sigma_A(s) = D(s) \sqrt{\Sigma_P / \Sigma_N}, \quad (12)$$

which is Read's σ_A with Luzzati's D . The per-atom model contains the standard likelihood as a special case with no approximation beyond (A1)–(A5), all of which the classical derivation also makes.

6 Spatial constraint

The per-atom model has $7N$ error parameters (w_j, U_j) . Amplitude data cannot support that many, and in the first moment they are exactly degenerate with occupancies (through w_j) and ADPs (through U_j added to U_j^{ADP}). The constraint adopted is to make the parameters smooth functions of position:

$$w_j = \exp(\bar{\lambda} + \lambda(x_j)), \quad U_j = U_0 + \frac{\kappa(x_j)}{8\pi^2} \not\ll, \quad (13)$$

where $\lambda(x)$ is the log presence contrast and $\kappa(x)$ the excess error-B, both mean-zero, band-limited (typically 15–20 Å) and symmetry-adapted; $\bar{\lambda}$ and U_0 set the spatial mean and are absorbed into the shell table $D_0(s)$. Combining, each atom’s effective weight becomes

$$d_j(h) = D_0(s) \exp(\lambda(x_j) - \kappa(x_j) s^2/4). \quad (14)$$

Equations (3)–(9) are unchanged; only the parameter count drops from $7N$ to a few dozen field coefficients plus the shell table. The fields are evaluated at atomic positions and never applied as a product in real space, because e^λ is not band-limited even when λ is.

Why the degeneracy with ADPs and occupancy is broken. In the mean (3), κ is indistinguishable from an ADP increment and λ from an occupancy. In the variance (4) they are not: a modelled ADP puts no variance on the Bragg reflections (physical motion sends its fluctuation into diffuse scattering), whereas the error field must pay $f_j^2 T_j^2 (w_j - d_j^2)$ of variance for every unit of mean it removes. Identifiability therefore rests on Σ_Δ being computed from the same (w_j, U_j) as F_{eff} . A freely refined per-shell variance would sever this link and reintroduce the degeneracy.

7 Regularisation

Write each field as a truncated, symmetry-adapted Fourier series with coefficients c_k , $c_0 = 0$. The full objective is

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \frac{1}{2} \sum_k \frac{|c_k|^2}{v_k} + \alpha \sum_s n_s \text{KL}(p_j(s) \parallel \text{uniform}). \quad (15)$$

The second term is a spectral taper (a stationary Gaussian process prior, diagonal in the Fourier basis on the periodic cell) with v_k decaying in $|k|$; it controls smoothness. The third is the relative entropy of the explained-power distribution in shell s ,

$$p_j(s) = \frac{d_j(s)^2 f_j^2 T_j^2}{\sum_i d_i(s)^2 f_i^2 T_i^2}, \quad (16)$$

weighted by the number of reflections n_s in the shell; it controls amplitude and favours down-weighting over up-weighting. Both terms are minimised by the uniform field, so regularisation pulls toward Section 5. The hyperparameters v_k and α are set by evidence maximisation.

8 Limit of a good model and strong data

Let the model be good: $w_j \rightarrow 1$, U_j small, $\Sigma_{\text{miss}} \rightarrow 0$. Expanding $d_j^2 = w_j^2 \exp(-4\pi^2 h^\top U_j h)$ to first order in U_j ,

$$w_j - d_j^2 \approx w_j(1 - w_j) + 4\pi^2 w_j^2 h^\top U_j h, \quad (17)$$

so

$$\Sigma_{\Delta}(h) \approx \sum_j f_j^2 T_j^2 \left[w_j(1 - w_j) + 4\pi^2 h^{\top} U_j h \right]. \quad (18)$$

The variance is small and, at high resolution, dominated by the positional-error term growing as s^2 : Luzzati's picture of coordinate error.

When $|F_{\text{eff}}|^2 \gg \varepsilon \Sigma_{\Delta}$, use $I_0(z) \sim e^z / \sqrt{2\pi z}$ in (7). The Rice density tends to a Gaussian in the amplitude,

$$|F| \approx \mathcal{N}(|F_{\text{eff}}|, \varepsilon \Sigma_{\Delta}/2), \quad (19)$$

and by the delta method the intensity has

$$\text{Var}(I_h) \approx 2 |F_{\text{eff}}(h)|^2 \varepsilon \Sigma_{\Delta}(h). \quad (20)$$

The negative log-likelihood (9) then reduces to weighted least squares on intensities,

$$\mathcal{L}_{\text{data}} \approx \sum_h \frac{(I_h - |F_{\text{eff}}(h)|^2)^2}{4 |F_{\text{eff}}(h)|^2 \varepsilon \Sigma_{\Delta}(h)} + \text{const}, \quad (21)$$

with weights equal to the inverse model variance of each intensity. This is the maximum-likelihood weighting for a Gaussian residual and the familiar least-squares target. Once experimental error is added, its variance σ_I^2 joins the denominator. The only non-classical content is that Σ_{Δ} is spatially informed.

9 Fisher closure

In this limit the per-atom errors need not be fitted. Let θ collect the model parameters and $J_h = \partial |F_{\text{eff}}(h)|^2 / \partial \theta$. The Gauss–Newton normal matrix of (21),

$$M = \sum_h \frac{J_h J_h^{\top}}{\text{Var}(I_h)}, \quad (22)$$

is the observed Fisher information, and M^{-1} is the asymptotic covariance of θ . Its 3×3 diagonal block for the coordinates of atom j is an estimate of that atom's positional uncertainty, which is exactly what U_j was defined to be:

$$U_j \leftarrow [M^{-1}]_{x_j x_j}. \quad (23)$$

This is the Cruickshank / Murshudov estimated-standard-uncertainty argument read in the opposite direction. It closes the model: the data determine U_j ; U_j determines Σ_{Δ} through (18); Σ_{Δ} sets the weights in (21); the weights determine M . Refinement and error estimation become a fixed-point iteration.

This fixes the role of the spatial fields. In the good-data limit the per-atom error model is computed from the Fisher information, not fitted, and the fields are unnecessary. When the data cannot support that (low resolution, partial or wrong models, $w_j < 1$, the Gaussian limit of (7) failing), the fields are the constrained interpolation between the Fisher answer and Read's uniform σ_A . The regulariser of Section 7 should therefore take the Fisher-derived U_j as its default, with the field carrying only what the data support beyond it.

10 Per-atom gradients

The target (9) depends on the atomic parameters through the pair $(F_{\text{eff}}, \Sigma_{\Delta})$, so by the chain rule

$$\frac{\partial \mathcal{L}}{\partial \theta} = \sum_h 2 \operatorname{Re} \left[\frac{\partial \mathcal{L}}{\partial F_{\text{eff}}^*(h)} \frac{\partial F_{\text{eff}}^*(h)}{\partial \theta} \right] + \sum_h \frac{\partial \mathcal{L}}{\partial \Sigma_{\Delta}(h)} \frac{\partial \Sigma_{\Delta}(h)}{\partial \theta}. \quad (24)$$

The two per-reflection factors $\partial \mathcal{L} / \partial F_{\text{eff}}^*$ and $\partial \mathcal{L} / \partial \Sigma_{\Delta}$ follow from (7)–(8) and are scalars; the first is the familiar m -weighted difference coefficient. The work is in the two model derivatives.

Mean term: Agarwal–Ten Eyck on a modified model. Atom j ’s contribution to F_{eff} in (3) is

$$d_j(h) f_j T_j e^{2\pi i h \cdot x_j} = D_0(s) w_j f_j \exp(-2\pi^2 h^{\top} (U_j^{\text{ADP}} + U_j) h) e^{2\pi i h \cdot x_j}, \quad (25)$$

so F_{eff} is exactly F_c of a *modified model* with occupancy w_j , ADP $U_j^{\text{ADP}} + U_j$, and an atom-independent resolution scale $D_0(s)$. Gradients of the mean term with respect to coordinates, occupancies and ADPs are therefore obtained by the standard FFT method (Agarwal 1978; Ten Eyck):

- (a) form the coefficients $D_0(s) \partial \mathcal{L} / \partial F_{\text{eff}}^*(h)$ (the scale D_0 is h -dependent but atom-independent, so it belongs in the map coefficients);
- (b) inverse-transform them to a gradient map $G(x)$;
- (c) sample G against the modified atomic density $\tilde{\rho}_j$ and its derivatives: $\nabla \tilde{\rho}_j$ for coordinates, $\tilde{\rho}_j$ for occupancy (hence for w_j), and $\partial \tilde{\rho}_j / \partial U$ for ADPs (hence for U_j).

The only practical difference from the usual case is that the effective ADP is inflated by U_j , which broadens the atoms and eases the sampling.

Variance term: no coordinate dependence. Σ_{Δ} in (4) contains no phase factors, so

$$\frac{\partial \Sigma_{\Delta}(h)}{\partial x_j} = 0. \quad (26)$$

For the coordinate cycle the Agarwal–Ten Eyck step is therefore sufficient by itself. The variance does depend on w_j , U_j and on T_j (the atom’s own ADP). Its gradients are phase-free sums:

$$\left. \frac{\partial \mathcal{L}}{\partial w_j} \right|_{\Sigma} = \sum_h \frac{\partial \mathcal{L}}{\partial \Sigma_{\Delta}(h)} f_j^2 T_j^2 (1 - 2w_j D_j(h)^2), \quad (27)$$

$$\left. \frac{\partial \mathcal{L}}{\partial U_j} \right|_{\Sigma} = \sum_h \frac{\partial \mathcal{L}}{\partial \Sigma_{\Delta}(h)} f_j^2 T_j^2 w_j^2 D_j(h)^2 4\pi^2 h h^{\top}. \quad (28)$$

Two evaluation routes:

- *Isotropic errors.* Each sum depends on atom j only through its element type and its total B . Bin $\partial \mathcal{L} / \partial \Sigma_{\Delta}$ radially once, then tabulate the sums per (type, B) on a one-dimensional grid; the per-atom cost is a lookup.

- *Anisotropic errors.* Define the variance-gradient map $A(x) = \sum_h (\partial\mathcal{L}/\partial\Sigma_\Delta(h)) e^{2\pi i h \cdot x}$. A sum of the form $\sum_h (\partial\mathcal{L}/\partial\Sigma_\Delta) f_j^2 T_j^2 D_j^2$ is then A integrated against the self-convolution (self-Patterson) of the modified atom, a Gaussian centred at the origin. This is the Agarwal sampling step with the atomic self-convolution evaluated at $x = 0$ instead of the atomic density at x_j : one extra FFT for the whole model.

Field coefficients. With the constraint (13), $\lambda_j = \lambda(x_j)$ and $\kappa_j = \kappa(x_j)$, and for a Fourier coefficient c_k of λ

$$\frac{\partial\mathcal{L}}{\partial c_k} = \sum_j \frac{\partial\mathcal{L}}{\partial \lambda_j} e^{2\pi i k \cdot x_j}, \quad (29)$$

and likewise for κ . Here $\partial\mathcal{L}/\partial\lambda_j$ combines the occupancy gradient of the mean term with (27), since $w_j = e^{\bar{\lambda} + \lambda_j}$; $\partial\mathcal{L}/\partial\kappa_j$ combines the ADP gradient of the mean term with the trace part of (28). The outer sum in (29) is a low-resolution structure-factor calculation with per-atom gradient weights in place of scattering factors: direct summation for a few dozen coefficients, a non-uniform FFT if the count grows.

Normal matrix for the Fisher closure. The FFT method also yields the sparse approximate normal matrix (Agarwal 1978; as used in REFMAC), whose 3×3 diagonal coordinate blocks are the blocks required by the closure (23). Off-diagonal blocks between neighbouring atoms affect the inverse and are handled by the usual sparse-inverse approximations.

Alternation. Since (26) holds, the natural schedule is: coordinates and ADPs refined against (21) or (9) with (w_j, U_j) frozen as constants; then (w_j, U_j) , or the field coefficients that generate them, refined with the atoms frozen. Evaluating $\lambda(x_j)$ live during the coordinate step would add a spurious force along $\nabla\lambda$ and is avoided.

11 Numerical checks required before implementation

Each item is to become a named computational artefact.

1. **Circularity (A4).** Draw many realisations of (1) for a synthetic model with known (w_j, U_j) ; estimate the pseudo-covariance of $F - F_{\text{eff}}$; confirm it is $O(\sqrt{N})$ relative to Σ_Δ and study its behaviour near special positions.
2. **Gaussianity (A3).** Test normality of $F - F_{\text{eff}}$ as a function of N and of the spread in f_j .
3. **Uniform limit (Section 5).** With constant (w, U) , confirm that (3)–(9) reproduce a standard σ_A fit to numerical tolerance.
4. **Least-squares limit (Section 8).** Confirm that (9) and (21) agree when $|F_{\text{eff}}|^2/\varepsilon\Sigma_\Delta \gg 1$, and that (20) matches the empirical intensity variance of synthetic draws.
5. **Fisher closure (Section 9).** On a noise-free synthetic crystal with known generating U_j , iterate (21)–(23) and confirm convergence to the generating values.

6. **Field recovery (Section 6).** Perturb one domain (displaced, partly deleted, error-inflated) and confirm that λ , κ and Σ_{miss} respond in the correct region and remain flat on a uniform-error control.
7. **Gradients (Section 10).** Finite-difference the analytic gradient on a small synthetic model, separately for the mean term, the variance term and the field coefficients; confirm (26) holds to machine precision.
8. **Null calibration.** Parametric bootstrap under the uniform field to obtain the distribution of the likelihood gain from a free field; use it as the acceptance threshold.